AI Data Engineer

Transition Roadmap

v2 — Validated & Updated May 2026

Boomi / UKG Developer → AI-Focused Data Engineering | 3-Month Sprint

Current Role	Target Role	Timeline	Firm
Boomi Developer	**AI Data Engineer**	**3 Months**	**Deloitte**

v2 Validation Changes (May 2026)

- FIXED: DataLemur & Interview Query replaced by DataDriven.io (100% free, no paywall, 1,400+ problems)
- ADDED: Git & GitHub (Week 1) — commit to GitHub from Day 1
- ADDED: Docker & containers (Week 5) — all DE tools run in Docker
- ADDED: Terraform / IaC (Week 6) — expected in cloud DE roles at Deloitte level
- ADDED: Apache Iceberg & Delta Lake (Week 7) — 78% adoption in new lakehouse deployments
- ADDED: Modern ingestion — Airbyte & dlt (Week 8) — replaced hand-rolled ETL in most stacks
- ADDED: Data Observability (Week 11) — Gartner forecasts 50% enterprise adoption by end 2026
- UPDATED: Plan expanded to 14 weeks to accommodate new topics without rushing

Month 1	Month 2	Month 3
Python · SQL · Git Cloud · First pipeline	Docker · Terraform · Airflow dbt · Iceberg · Airbyte Spark · Kafka	Data Observability · RAG ML pipelines · Vector DBs Capstone · Cert

Section 1: Your Boomi Advantage

You are not starting from zero. Your Boomi + UKG WFM background has built real data engineering instincts that most candidates lack.

Your Boomi Skill	Data Engineering Equivalent
• Boomi processes & shapes • API/UKG connectors • Map functions & data transformation • Boomi Flow & process scheduling • Error handling & retry logic • EDI / flat file processing	• Airflow DAGs / Prefect flows • REST API ingestion & Kafka consumers • dbt models / PySpark transformations • Cron + Airflow schedulers / Lambda triggers • Dead letter queues, alerting & circuit breakers • CSV / JSON / Parquet ingestion pipelines

Key Insight

- On your resume and in interviews, reframe every Boomi pipeline as an ETL/ELT pipeline. You have already designed data flows, handled transformations, managed error states, and ensured SLAs — that is precisely what data engineers do.

What You Already Have vs What You Need to Add

What You Already Have	What You Need to Add
• Understands data movement and integration • Built error handling & retry patterns • Worked with APIs and flat files at scale • Understands scheduling & dependency management • Real Deloitte client delivery experience	• DE tooling: Docker, Airflow, dbt, Spark, Kafka • Open table formats: Iceberg & Delta Lake • Modern ingestion: Airbyte / dlt • Cloud data platforms (BigQuery, Redshift) • AI / LLM data pipeline skills (RAG, vector DBs) • GitHub portfolio + cloud certification

MONTH 1 — Foundations (Weeks 1–4)

Goal: Git fundamentals, Python & SQL at a data engineering level, cloud basics, and your first end-to-end pipeline.

Git & GitHub — Start Week 1

Git essentials to learn in Week 1
• git init, clone, add, commit, push, pull — the daily workflow • Branching: feature branches, pull requests, merge • GitHub: create a profile, pin repositories, write a README for every project • Resource: skills.github.com (free interactive) + docs.github.com/get-started

Python Refresher — Data Focus

Core Python Skills	Advanced / Pipeline-Specific
• Pandas: DataFrames, groupby, merge, pivot, apply	• Working with APIs (requests library)
• NumPy basics for array operations	• Pydantic for data validation
• File I/O: CSV, JSON, Parquet, Avro	• asyncio basics for async pipelines
• List comprehensions, generators, decorators	• Pytest for pipeline testing
• Virtual environments, pip, requirements.txt	• Environment variables & .env files (dotenv)
• Error handling: try/except, logging module	• Packaging scripts as reusable modules

SQL — Advanced Patterns

Advanced SQL Concepts	DE-Specific Patterns
• Window functions: ROW_NUMBER, RANK, LAG, LEAD	• Slowly Changing Dimensions (SCD Type 1 & 2)
• CTEs and recursive queries	• Dimensional modelling: star & snowflake schemas
• Query optimization — EXPLAIN, indexes	• MERGE / UPSERT patterns for incremental loads
• JSON/ARRAY functions (Postgres, BigQuery)	• Analytical queries on HR / workforce data
• Partitioning and clustering strategies	• Data quality checks via SQL assertions

Cloud Fundamentals — Pick One Platform

AWS (Widest Job Market)	GCP (Best for Data / AI Work)
• S3: object storage, lifecycle policies	• BigQuery: serverless warehouse (recommended)

• Glue: serverless ETL (closest to Boomi) • Redshift: data warehouse • Lambda: event-driven Python functions • IAM: roles, policies, least privilege	• Cloud Storage (GCS): data lake • Cloud Composer: managed Airflow • Pub/Sub: event streaming • Vertex AI: ML platform integration

Week-by-Week Schedule — Month 1

Week	Focus Area	Topics	Deliverable
W1	**Git + Python**	• Git: commit, branch, push to GitHub every day • Pandas: merge, groupby, reshape, apply • File I/O: CSV, JSON, Parquet	*GitHub repo + Python transform script*
W2	**Python + APIs**	• requests: REST API calls with auth • Pydantic validation, error handling • Logging, dotenv, argparse for CLI tools	*Script: pull data from a public API*
W3	**Advanced SQL**	• Window functions — DataDriven.io daily • CTEs, recursive queries, MERGE patterns • Queries in BigQuery or Redshift free tier	*SQL portfolio: 10 problems solved*
W4	**Cloud + Pipeline**	• Create S3 / GCS bucket, set up IAM • Python script: load data to cloud storage • Schedule with Lambda / Cloud Function	*API → transform → cloud storage*

MONTH 2 — Core Data Engineering (Weeks 5–10)

Goal: Docker, Terraform, orchestration, open table formats, modern ingestion, big data and streaming.

Docker & Containers — Week 5 (NEW ✓)

Docker essentials for data engineers

- Containers vs VMs — why every DE tool runs in a container
- Dockerfile: containerise your Python pipeline script
- Docker Compose: run multi-service stacks (Postgres + pgvector, Kafka + Zookeeper)
- Key commands: docker build, run, ps, exec, logs, docker-compose up / down
- Resource: docs.docker.com/get-started (free official tutorial, ~4 hrs)

Terraform — Infrastructure as Code (NEW ✓)

Terraform basics for data engineers

- What IaC means: define cloud resources (S3, BigQuery datasets) as reproducible code
- Core workflow: providers, resources, state, terraform plan, apply, destroy
- Write a Terraform config to create an S3 bucket and Glue crawler
- Resource: developer.hashicorp.com/terraform/tutorials (free, browser-based labs)

Airflow — Orchestration Engine

Core Airflow Concepts	Advanced Patterns
• DAG structure: tasks, dependencies, schedule • Operators: PythonOperator, BashOperator • Hooks: connecting to databases, S3, GCS • XComs: passing data between tasks • SLAs, retries, alerting (email / Slack)	• TaskGroups for complex DAG organisation • Dynamic DAGs with task mapping • Connections and Variables (secrets management) • Airflow 3 features (new 2025 release) • Monitoring: Airflow UI, logs, Flower

dbt — Data Transformation Layer

dbt Fundamentals	Advanced dbt Patterns
• Project structure: models, seeds, tests, macros • Staging → intermediate → mart layer pattern • ref() and source() for dependency tracking • Schema tests: not_null, unique, accepted_values	• Snapshots for SCD Type 2 history tracking • Macros and Jinja templating • dbt documentation and lineage graphs • dbt Cloud free tier for dev and deploy

• Incremental models for large tables	• dbt + Airflow via Astronomer Cosmos

Apache Iceberg & Delta Lake (NEW ✓)

Why open table formats matter — 78% adoption in new lakehouses (2026)
• Problem solved: traditional data lakes had no ACID transactions, schema evolution, or time travel • Apache Iceberg: ACID transactions, hidden partitioning, schema evolution on S3/GCS • Delta Lake: Databricks' open format — you already use it in Spark on Databricks Community Edition • Time travel: query data as it was at a past point — critical for audit and ML pipelines • Apache Hudi: third option used at Uber — good to know conceptually • Resources: iceberg.apache.org/docs (free) + datalakehousehub.com beginner guides (free)

Modern Ingestion — Airbyte & dlt (NEW ✓)

Airbyte	dlt (data load tool)
• Airbyte: open-source EL(T) with 300+ connectors • Run Airbyte locally via Docker Compose • Connect sources to destinations (BigQuery, S3) • Airbyte Cloud free tier for experimentation	• dlt (data load tool): Python-native lightweight ingestion • pip install dlt — no server, no YAML config • Write ingestion pipelines as pure Python code • dlt + dbt: the modern lightweight stack alternative

Spark & Kafka

Apache Spark (PySpark)	Kafka — Streaming Basics
• PySpark DataFrames, joins, window functions • Reading / writing Parquet, Delta Lake, Iceberg • Partitioning strategy and avoiding data skew • Run on Databricks Community Edition (free)	• Kafka: topics, partitions, offsets, consumer groups • Producers and consumers in Python • Kafka + Spark Structured Streaming • Schema Registry and Avro serialisation

Week-by-Week Schedule — Month 2

Week	Focus Area	Topics	Deliverable
W5	**Docker**	• Install Docker Desktop, learn key commands • Docker Compose: Postgres + pgvector locally • Containerise your Month 1 Python pipeline	*Docker Compose stack running locally*

W6	**Terraform + Airflow**	• Terraform: create S3/GCS bucket as code • Astro CLI: build first 3 DAGs locally • Connect Airflow to cloud storage via Hooks	*Terraform config + Airflow DAGs running*
W7	**dbt + Iceberg**	• dbt init, staging/mart layers, tests, ref() • Explore Delta Lake tables on Databricks • Read Iceberg: schema evolution + time travel	*dbt project on BigQuery + Delta table*
W8	**Airbyte + dlt**	• Run Airbyte on Docker, create a connector • pip install dlt, write ingestion pipeline • Compare Airbyte vs dlt for use cases	*Airbyte pipeline ingesting to BigQuery*
W9	**Spark (PySpark)**	• Databricks Community Edition setup • DataFrames: read, transform, write Parquet + Delta • Joins, window functions, aggregations at scale	*Spark job: 1M+ rows on Databricks*
W10	**Kafka + Integration**	• Kafka producer/consumer in Python on Docker • Chain Airflow → dbt → Spark end-to-end • Add monitoring, logging, alerting throughout	*Full stack: Airflow+Airbyte+dbt+Spark+Kafka*

MONTH 3 — AI Data Engineering + Portfolio (Weeks 11–14)

Goal: Data observability, AI-specific data infrastructure, a GitHub portfolio, and your cloud certification.

Data Observability (NEW ✓)

The 5 pillars — Gartner forecasts 50% enterprise adoption by end of 2026

- Freshness: is your data up to date? Monitor table update cadence and alert on staleness
- Volume: are the expected number of rows arriving? Flag drops or spikes automatically
- Distribution: are field values within expected ranges? Catch data drift before it hits dashboards
- Schema: have column types or names changed unexpectedly? Break pipelines before bad data spreads
- Lineage: track how data flows from source → transformation → destination
- Tools to learn: Great Expectations (free, pip install) + dbt tests + Soda Core (free, open source)
- Concepts to know: Monte Carlo (commercial leader) + OpenLineage (open standard for lineage)

ML Pipeline Infrastructure

ML Data Infrastructure	Advanced MLOps Patterns
• Feature engineering: lag features, rolling windows	• Model serving data pipelines (batch prediction)
• Feature stores: Feast (open source) concepts	• Data drift detection: evidently, deepchecks
• Data versioning with DVC (Data Version Control)	• Retraining triggers based on data quality metrics
• MLflow: experiment tracking and model registry	• A/B test data logging and analysis pipelines

LLM & Generative AI Data Infrastructure

LLM Data Engineering Stack

- Vector databases: Pinecone, Weaviate, pgvector (Postgres extension) — store and retrieve embeddings
- Embedding pipelines: chunk documents → embed via OpenAI/Cohere API → store in vector DB
- RAG (Retrieval-Augmented Generation): the dominant LLM app architecture you will support
- LangChain data loaders: ingest PDFs, web pages, databases into LLM-ready format
- LLM evaluation pipelines: log prompts, completions, latency, token usage
- Real-time vs batch embedding refresh strategies for production RAG systems

Capstone Project — Full v2 Stack

Project: AI-Powered HR Analytics Pipeline (updated stack)

- Ingest: Airbyte pulls Kaggle HR dataset into GCS/S3, orchestrated by Airflow
- Store raw: Apache Iceberg tables on object storage for ACID + time travel
- Transform: dbt models — staging → HR analytics marts (attrition, headcount)
- Observe: Great Expectations + dbt tests — data quality at every layer
- Enrich: OpenAI embeddings API on HR survey free-text responses
- Store vectors: pgvector (Postgres on Docker) for semantic search
- Serve: Streamlit UI — HR asks natural language questions over the data
- Infrastructure: Docker Compose for local dev, Terraform for cloud provisioning

Week-by-Week Schedule — Month 3

Week	Focus Area	Topics	Deliverable
W11	**Data Observability**	• Great Expectations: write pipeline expectations • dbt tests: custom SQL assertions on marts • OpenLineage concepts + Airflow integration	*Fully tested pipeline — GE + dbt quality*
W12	**ML + RAG**	• MLflow: log experiments, compare, register model • Embed documents via OpenAI API, store pgvector • Build basic RAG pipeline with LangChain	*MLflow tracking + RAG: PDFs → embed → query*
W13	**Capstone Build**	• Wire: Airbyte + Airflow + dbt + Iceberg + Spark • Data quality + observability layer fully active • Streamlit UI for natural language HR queries	*Capstone on GitHub with README + demo URL*
W14	**Cert + Job Ready**	• AWS/GCP Data Engineer Associate exam • Update resume with all new DE/AI skills • Connect with Deloitte A&C / AI & Data practice	*1 cert + 2 GitHub projects + updated resume*

Section 5: Validated Resources & Tools

Python, Git & SQL

Resource	Type	Time	Link / Platform
Kaggle Python Course	Free	5 hrs	kaggle.com/learn/python
Kaggle Pandas Course	Free	4 hrs	kaggle.com/learn/pandas
GitHub Skills	Free Interactive	2 hrs	skills.github.com
DataDriven.io — SQL + Python	100% Free	Ongoing	datadriven.io (no paywall ever)
LeetCode SQL 50	Free	Ongoing	leetcode.com/studyplan/top-sql-50
Mode SQL Tutorial	Free	6 hrs	mode.com/sql-tutorial

Data Engineering Core

Resource	Type	Time	Link / Platform
Docker Get Started	Free Official	4 hrs	docs.docker.com/get-started
Terraform Tutorials	Free Interactive	3 hrs	developer.hashicorp.com/terraform/tutorials
dbt Fundamentals	Free Course	6 hrs	courses.getdbt.com
Astronomer Academy (Airflow 3)	Free Courses	8 hrs	academy.astronomer.io
Databricks Community Edition	Free Platform	Ongoing	community.cloud.databricks.com
DataTalks.Club DE Zoomcamp	Free Cohort	9 wks	github.com/DataTalksClub/data-engineering-zoomcamp
Airbyte Docs	Free/Open Source	Ongoing	docs.airbyte.com
dlt (data load tool) Docs	Free/Open Source	Ongoing	dlthub.com/docs
Apache Iceberg Docs + Hub	Free	Ongoing	iceberg.apache.org/docs + datalakehousehub.com

AI / LLM Data Engineering

Resource	Type	Time	Link / Platform
LangChain Python Docs	Free Docs	Ongoing	python.langchain.com/docs
Pinecone Learning Center	Free Course	3 hrs	pinecone.io/learn
OpenAI Cookbook	Free Tutorials	Ongoing	cookbook.openai.com
Microsoft Gen AI for Beginners	Free 18-lesson	Self-paced	github.com/microsoft/generative-ai-for-beginners
DataTalks.Club LLM Zoomcamp	Free Cohort	Ongoing	github.com/DataTalksClub/llm-zoomcamp
Great Expectations Docs	Free/Open Source	Ongoing	docs.greatexpectations.io

Soda Core Docs	Free/Open Source	Ongoing	docs.soda.io
MLflow Quickstart	Free Docs	2 hrs	mlflow.org/docs/latest/quickstart

Certifications — Target Month 3

Resource	Type	Time	Link / Platform
AWS Certified Data Engineer – Associate	Cert	4-6 wks	aws.amazon.com/certification
GCP Professional Data Engineer	Cert	6-8 wks	cloud.google.com/learn/certification
Databricks Associate Developer for Spark	Cert	3-4 wks	databricks.com/learn/certification
dbt Analytics Engineering Cert	Cert	1-2 wks	credentials.getdbt.com

Section 6: Deloitte Internal Strategy

An internal transfer at Deloitte is often faster and more credible than an external search. Here is how to engineer that move.

Target Practices

Where to Look	What to Search For
• Analytics & Cognitive (A&C) — primary target • AI & Data practice • Cloud Engineering (AWS/Azure/GCP data work) • Government & Public Sector data teams	• Search: Spark, Airflow, dbt, BigQuery, Iceberg, LLM, data pipeline • Titles: Data Engineer, AI Engineer, MLOps Engineer, Data Platform Engineer • Ask your counsellor about rotational / secondment opportunities

Month-by-Month Internal Moves

- Month 1–2: Find a project where data migration or pipeline modernisation is in scope and volunteer. On-the-job learning with real client exposure.
- Month 2–3: Propose a Boomi → Airflow / cloud-native migration for part of the UKG WFM integration. Shows initiative and technical growth to leadership.
- Month 3: Request an informational chat with A&C or AI & Data practice. Ask how they staff projects and what skills are most in demand right now.
- Month 3: Update your Deloitte profile with new skills (Spark, dbt, Airflow, Docker, Iceberg, Airbyte, Python, vector databases) and certifications.
- Throughout: Connect with Deloitte's AI Centre of Excellence (CoE) on Teams. Attend internal tech talks, innovation days, and AI community events.

Resume Reframing — Critical

Before (Boomi framing)	After (DE/AI framing)
• "Designed and implemented Boomi integration processes for UKG WFM to SAP"	• "Architected and deployed end-to-end ETL pipelines integrating UKG WFM workforce data with SAP, processing [X]K records daily with automated error handling, retry logic, and SLA alerting"

Section 7: Success Checklist — v2

Month 1 Milestones

- [ ] GitHub repo live, committing code every week from Day 1
- [ ] Python scripts for data ingestion and transformation working locally
- [ ] Advanced SQL: 10+ problems solved on DataDriven.io or LeetCode
- [ ] Cloud account set up (AWS free tier or GCP $300 credit)
- [ ] First pipeline: API → Python transform → cloud storage

Month 2 Milestones

- [ ] Docker: multi-container stack running locally (Postgres + your pipeline)
- [ ] Terraform: config created and applied to provision a cloud resource
- [ ] Airflow: Astro CLI running locally, 3 working DAGs built
- [ ] dbt project with staging + mart models and all tests passing
- [ ] Apache Iceberg / Delta Lake: reading and writing tables on Databricks
- [ ] Airbyte or dlt: ingestion pipeline pulling from a real source to cloud
- [ ] PySpark job on Databricks processing a real dataset (1M+ rows)
- [ ] Kafka: producer and consumer running on Docker
- [ ] Started studying for AWS / GCP Data Engineer certification

Month 3 Milestones

- [ ] Data quality: Great Expectations + dbt tests passing on all pipeline layers
- [ ] RAG pipeline: documents → embeddings → pgvector → queried
- [ ] MLflow: experiment tracking integrated into a pipeline
- [ ] Capstone project on GitHub (README + Streamlit demo link)
- [ ] AWS Certified Data Engineer – Associate OR GCP Professional DE cert earned
- [ ] Resume updated with full DE/AI stack (Docker, Iceberg, Airbyte, RAG, etc.)
- [ ] LinkedIn updated: new skills, certifications, GitHub link prominent
- [ ] At least 1 conversation with Deloitte A&C / AI & Data practice member
- [ ] Job applications submitted (internal at Deloitte and/or external)

Final Note

- The transition from integration developer to AI data engineer is one of the most logical career moves in tech right now. Your Boomi background means you already understand data movement, transformation, and integration at a fundamental level. The v2 roadmap now reflects exactly what the 2026 market expects — Docker, Terraform, Iceberg, Airbyte, and data observability included. Three to four months of deliberate, consistent effort will put you in a strong position. Start this week.